

# Industrialized Riverine Landscapes - How extremes, infrastructure, and resource demands transform socio-ecohydrological systems

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**Doctoral Programme:**

**BOKU Doctoral School Human River  
Systems in the 21<sup>st</sup> Century (HR21)**

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# 1. Research Framework

## 1.1. Introduction - the coupled socio-ecohydrological evolution of rivers<sup>1</sup>

Since the onset of the Anthropocene, landscapes and river systems have undergone dramatic, worldwide transformation. This transformation was driven by a shift from an agrarian (solar-energy-based) to an industrialized (fossil-fuel-based) socio-metabolic regime <sup>[2]</sup> with impacts on the availability and quality of water and the functioning of aquatic ecosystems. Consequently, almost all river systems can no longer be considered as natural systems. Dynamic, complex interactions and feedback mechanisms between natural and social systems characterise today's rivers and their catchments, they have been transformed into **IRL – Industrialized Riverine Landscapes**.

Awareness of these interactions has increased remarkably; the initiative of the hydrological community announcing the Scientific Decade of Panta Rhei 2013-2022 is only one case in point <sup>[3]</sup>. Concepts and research questions to target linkages of hydrological and societal aspects were developed <sup>[3]</sup> addressing urgent unsolved problems as highlighted in several joint publications <sup>[4,5]</sup>. There is a strong need to go beyond social-hydrological interactions <sup>[6]</sup> by integrating **socioeconomic, ecological, and hydrological perspectives** on an **equal footing** in a **novel research framework** <sup>[1]</sup>.

Based on an interdisciplinary research driven education, this Doctoral School studies IRLs as coupled **socio-ecohydrological systems (SEHS)** <sup>[1]</sup>, asking how they respond to current and future environmental, social, cultural, and economic drivers of change, including climate, demographic, and land-use change. This Doctoral School trains a **new generation of experts** and provides scientific evidence for sound adaptive management supporting alternative transformation pathways of IRL into self-sustaining and more resilient systems. This Doctoral School embodies the visions of the University of Natural Resources and Life Sciences, Vienna (BOKU), as a leading university for sustainability in Europe. The faculty involved in supervising and educating the doctoral students stands out for their expertise in natural, engineering, and social sciences and their high potential to support **evidence-based adaptive management**.

## 1.2. Challenges for science and society demanding integrated approaches

The **industrialization of riverine landscapes** has **altered aquatic ecosystems** and created complex interactions with societal activities, leading to compound pressures from which aquatic systems can only slowly (or never) recover, and which impair the provision of services from aquatic ecosystems <sup>[7, 8]</sup>. These pressures can have severe impacts and can lead to, for example, a loss of biodiversity <sup>[9]</sup> soil sealment by settlements and infrastructure <sup>[10]</sup>, deterioration of water quality <sup>[11]</sup>, and to imbalances in water security <sup>[12]</sup> and sediment dynamics <sup>[14,15]</sup> in rivers and their catchments worldwide.

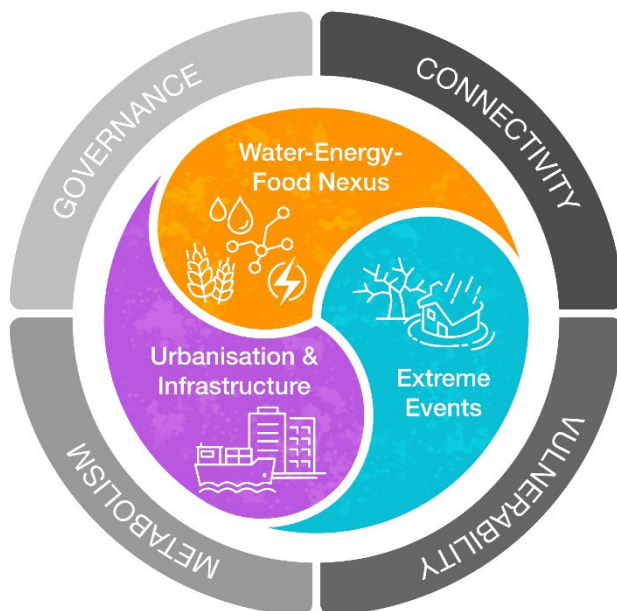
Fossil-fuel driven industrialization <sup>[16]</sup> has led to an unfolding climate crisis, accompanied by more **extreme events** of floods and droughts, <sup>[17]</sup> rapid **urbanisation and infrastructure** development and **overexploitation of natural resources**. This is challenging the **water-energy-food nexus** which in turn impairs the ecological and chemical status of aquatic ecosystems <sup>[18,19]</sup>. This transformation into IRL decreased rivers' self-purification capacity, provision of ecosystem services, and biodiversity refuge, which in turn increases the vulnerability of ecological as well as socio-economic systems and thus puts current and future generations at risk <sup>[20]</sup>. Many of these human-induced pressures on the environment are persistent, highly complex, sometimes intertwined, often global, and with delayed responses, with extensive spatial and temporal distance between cause and effect, usually involving numerous conflicting stakeholders <sup>[21]</sup>. IRL not only need to be adapted

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<sup>1</sup> The research framework outlined in chapter 1 of this proposal is the result of common conceptual work over the last years in the already established BOKU doctoral school HR21; it has been published in more detail in a joint publication co-authored by PhD-students and faculty <sup>[1]</sup>.

to global challenges, such as climate change, they are also part of potential solutions, e.g., through changes in carbon emissions, savings, or sequestration [22, 23].

Modern societies appear ill-equipped to cope with the drastic modifications and **accelerated rates of change** in IRLs. While they depend on their integrity and ecological as well as societal functionality, they are faced with the **urgent need for action** (e.g., land-use change, sediment management, climate mitigation and adaptation, freshwater biodiversity crisis, water availability, mitigation of infrastructure effects) and the concurrent need for long-term, sustainable management. Improved predictions as a basis for adaptive management and the assessment of the effectiveness of Nature-based-Solutions (NbS) are urgently needed to meet ecological, economic, and societal demands in the light of global change. Such challenges require interdisciplinary approaches, as multiple pressures are interacting at different temporal and spatial scales. Crucial issues to be managed in the future include the continued fast-growing societal resource demands that are addressed in the **water – energy – food nexus** (e.g. expansion of renewable energies), water availability impacted by **hydrologic extremes**, and the effects of **infrastructure development** on both, riverine processes and society. Innovation is needed, e.g., in understanding cascading effects of droughts and floods, the interaction of social and ecological metabolism, and how industrialization impacts river systems and society, also via related feedback mechanisms.



Within the existing doctoral school, a research framework (**HR21 framework**) composed of four research clusters (connectivity, vulnerability, metabolism, and governance) was developed (Fig. 1). Within this framework, we defined **three cross cutting research fields** to first address some of the most pressing challenges [24] in current IRL to be investigated in **15 PhD research topics**. Those topics are based on, and integrate, the key expertise of faculty members from different disciplines and therefore go beyond the state-of-the-art of an individual sub-discipline.

Figure 1: The three cross cutting research fields (indicated in orange, purple and cyan) embedded in our framework of four research clusters, based on Hein et al. 2021.

#### Box: Innovation

The Doctoral School HR21 targets real-world problems and goes beyond social-hydrological interactions by **integrating** for the first time **socioeconomic, ecological, and hydrological perspectives in its research framework**. Such research considers biophysical and socioeconomic processes equally and analyses the accelerating industrialization of riverine landscapes and its manifestations in form of (i) the climate crisis with more **extreme events** of floods and droughts, (ii) rapid **urbanisation and infrastructure development** and (iii) overexploitation of natural resources related to the **water-energy-food nexus**.

In the following, we first explain the overarching HR21 framework based on four research clusters (RC) that organize the interdisciplinary challenges related to IRL. In a next step, we detail the three research fields (RF) that will be interdisciplinary investigated by 15 PhDs by incorporating aspects of one or more of the four research clusters.

### 1.3. HR21 framework

Our approach is based on the **distinction between natural (eco-hydrological) and social (socio-cultural-economic) systems**, which we see as co-evolving SEHS <sup>[25]</sup>. The study of SEHS requires the expertise from three research domains: i.e. natural sciences, engineering, and the social sciences and humanities along with their integration and synthesis, embedded in **four research clusters (RC)**. Transformation processes in IRL are the result of coupled natural and social processes in river systems. Hence, understanding this coupling of eco-hydrological with social systems including relevant actors and institutions through the concepts of metabolism and colonization as defined in Social Ecology <sup>[26,27]</sup> is the foundation of our interdisciplinary research.

Our framework is structured into four RC: (i) **Connectivity**, addressing social, ecological, hydrological, and landscape spatial properties of IRL, (ii) **Governance** and planning addressing mainly how social systems interact with IRL, (iii) **Metabolism** of social and natural systems, addressing how societal flows of matter and energy interact with ecohydrological systems as sources and sinks, and (iv) **Vulnerability** to impacts, risks, and shifting baselines essentially influencing how ecohydrological and social systems co-evolve in SEHS.

#### **Connectivity**

IRL exhibit altered **connectivity properties** with new spatial arrangement of landscape characteristics (habitat types, land use, and riverine corridors) and exchange processes of water, matter and biota. Hydraulic engineering or soil sealing changed connectivity properties especially at local and regional scales <sup>[28]</sup>. For example, land use change in floodplains limits the availability of floodplain habitats for riverine communities and impacts floodwater retention and recreational uses. Following a meta-ecosystem approach <sup>[29]</sup>, we conceptualize and analyse cross-scaling downstream effects of such interventions to gain new insights in SEHS and identify **novel emergent properties** of the resulting industrialized system. Considering the structural and functional connectivity identified in **network analyses** will enable innovative ways to quantify changes in SEHS properties <sup>[30]</sup>.

#### **Governance and Planning**

To enhance resilience and diversity of IRL as living spaces for humans and other species, a major transformation of the appropriation of resources from IRL (soil, water, air, energy, biodiversity, landscape, etc.) must take place. This **calls for new strategic planning and governance framework** practices as well as for pro-active and re-active management processes promoting sustainability <sup>[31,32]</sup>. Therefore, this RC addresses the interplay of formal planning and governance- and stakeholder processes to manage IRL at different scales, across sectors, scientific disciplines, and administrative borders. It addresses questions of how policy making and adaptive management can be optimized to cope with different drivers of change and various human needs and, thus, promotes sustainable development of IRL.

#### **Metabolism**

This RC addresses rivers' changing role in society's metabolism <sup>[33]</sup> and how this is connected at smaller spatial scales, to metabolic processes in riverine landscapes and aquatic organisms <sup>[36]</sup> for a more sustainable coupling of social with ecohydrological systems in IRL. Rivers have been and are sources for indispensable material and energetic inputs into society, they provide space for infrastructure, and as sinks they receive output flows from the social system. IRL are major sources of energy for society, as transport routed and sites of electricity generation they are sources for our 'high energy civilizations' <sup>[35]</sup>. Fossil fuel-driven industrialization of the social metabolism has significantly changed the scale, intensity, and ecological legacies of human interventions, transforming rivers into IRL. A **socio-metabolic research approach** <sup>[34]</sup> is combined with research in river ecology and engineering to envisage more sustainable IRL and deepen understanding of the coupling of natural and social systems in SEHS.



## **Vulnerability**

Vulnerability is a function of exposure, sensitivity, and adaptive capacity. Assessing the vulnerability of aquatic ecosystems to (multiple) human pressures and climate change is key for informing water management and developing sustainable solutions for water security and ecosystem health. **Vulnerability is not only a component of risk assessment** <sup>[3]</sup> but has also been applied for the **integrated analysis of socio-economic and environmental aspects** in sustainability assessments <sup>[38,39]</sup>. In the past, static systems' states and the impacts of single pressure-response processes were assessed <sup>[40]</sup>. However, recent developments show that systems are highly dynamic, and superimposed by multiple pressures at the same time causing no clear baseline conditions, hardly predictable consequences for riverine processes, aquatic organisms and human society <sup>[41]</sup>.

Our research framework shares common scientific concerns related to **spatial and temporal scales, prediction, uncertainty and indicators**. Within our socio-ecohydrological framework, it is important to consider that interventions and processes take place at one location, while the impacts are observed elsewhere, and decisions are made in a third place. Linkages need to be addressed between spatial scales of processes or interventions and the organization and spatial competence of the decision-making institutions and authorities. The impacts can occur instantaneously on short temporal scales or they may be delayed severely due to water transit times, ranging from days to several hundreds of years along individual flow paths <sup>[42]</sup> as well as due to long-term alterations of biogeochemical processes (i.e., legacy effects) and societal adaptations. Insights from environmental history are helpful by considering the specific temporal horizons of political and socio-cultural processes, ecosystem services (ESS), and their long-term legacies in both society and nature <sup>[43]</sup>.

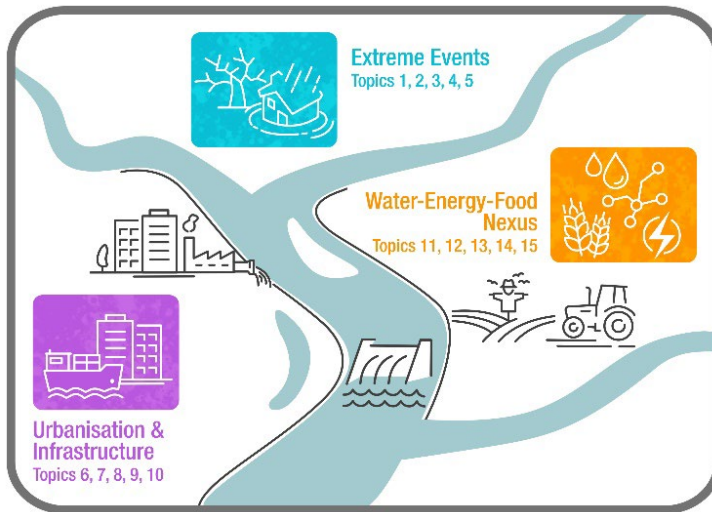
For improving integrated water management, we need to better understand the **consequences of interventions** and their different temporalities, to improve predictions and projections, and reduce their uncertainty. Understanding how SEHS react to drivers of change needs the integration of new methods and tools that allow for the simulation of processes and feedback mechanisms. The spectrum of methods includes the analysis of past governance schemes and ensuing conflicts, the development and application of coupled **socio-ecohydrological modelling tools** <sup>[43]</sup> of different complexity, also including statistical or machine learning approaches <sup>[45]</sup>. One of the key challenges is the consideration and quantification of the **full 'chain of uncertainty'** within the cycle of socio-economic scenarios relevant to IRL, their regional projections, subsequent impact analysis, the realization of adaptation measures, and possible feedbacks <sup>[46]</sup>. Improved predictions and their known and reduced uncertainty are the basis for **adaptive planning and management**. Assessment schemes and development of indicators have already advanced in the recent past, but **new indicators** are needed to improve the implementation of societal dynamics into water and land management <sup>[47]</sup>. The products of the PhD projects will also be new modelling tools, like **integrated modelling frameworks** and **decision support tools** (including stakeholder perspectives), to facilitate better informed management decisions in IRL.

While the four RC provide the conceptual foundation for HR21 via influential, interdisciplinary scientific approaches, three **cross-cutting research fields** (RF) in IRL guide our research via specific topics considered as crucial for the sustainable transformation of SEHS (cp. Figure 1).

### **1.4. Cross-cutting Research Fields (RF)**

To understand the transformation of IRL in the Anthropocene <sup>[48]</sup>, our research addresses major challenges related to climate and land-use change in urban and rural catchments and the resource demand of modern societies. Three major **cross-cutting research fields** (RF) will be investigated by the 15 PhD projects proposed: **(1) extreme events, (2) urbanization and infrastructure**, and

**(3) the water - energy - food nexus** (Fig. 2) and each topic relates to at least two RC (see Table 1, Attachment 1). Thus, the proposed research plan would allow **15 well-orchestrated joint PhD projects** starting at the same time targeting these key research fields, compared to individually designed PhD projects in the current doctoral programme. Each PhD topic will be finally designed in detail by students in a **PhD thesis concept** in the initial phase, allowing also to specify **collaborations within the doctoral school** and based on available data.



Together, the PhD topics, RC and RF reflect the **faculty's interdisciplinary expertise** and the broad spectrum of conceptual and methodological approaches assembled in HR21.

Through **cooperation with international partners**, the doctoral school further strengthens the research portfolio at BOKU and increases its visibility in integrated water research at the national and international level.

Figure 2: The 15 PhD topics within the three cross-cutting research fields and their spatial arrangement in IRL.

Tab. 1: Overview on PhD topics, their links to RF, RC, and interactions with other PhD topics.

| No. | PhD topic                                                                                                                                    | IRL Supervisor       | research field | research clusters | link to other PhD topics |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|-------------------|--------------------------|
| 1   | Role of floods for sediment transport and morphodynamics in the human-river system                                                           | Helmut Habersack     | Extremes       | Gov, Vul, Met     | 2, 6, 7, 8               |
| 2   | Material resources demand and flows of hydraulic structures to prevent societal impact of extreme events                                     | Gertrud Haidvogel    | Extremes       | Met, Gov          | 1, 6, 7, 8, 10           |
| 3   | Intermittent flow – hydrological effects and impacts on benthic communities, ecological state and functioning of riverine systems            | Wolfram Graf         | Extremes       | Vul, Con, Met     | 4, 5, 14                 |
| 4   | Drought, low flow and warming – hydrological effects and impacts on the ecological state and functioning of river systems                    | Gregor Laaha         | Extremes       | Vul, Con, Met     | 3, 5, 14                 |
| 5   | Water transit time distributions as indicators for the vulnerability of riverine ecosystems under different pressures from social metabolism | Christine Stumpp     | Extremes       | Con, Met, Vul     | 2, 3, 4, 9, 11, 12       |
| 6   | Analysing the contribution of re-activated former watercourses towards a more sustainable transformation of IRL                              | Thomas Ertl          | Urb_Infra      | Gov, Vul, Met     | 8, 9, 10, 15             |
| 7   | The role of the latest industrial revolutions on instream processes and ecological vulnerability of IRL                                      | Christoph Hauer      | Urb_Infra      | Vul, Met, Gov     | 8, 9, 10, 13, 1, 2       |
| 8   | The effects of infrastructure development on ecological and ecosystem service network properties in IRL                                      | Thomas Hein          | Urb_Infra      | Con, Met          | 6, 7, 10, 1, 2           |
| 9   | The impact of urban settlements on organic matter and nutrient flows in IRLs – What can we learn from the past?                              | Günter Langergraber  | Urb_Infra      | Met, Gov          | 6, 10, 11                |
| 10  | Urban metabolism, a river's ecohydrology and industrialisation: Vienna and the Danube, 1800-2000                                             | Martin Schmid        | Urb_Infra      | Met, Con          | 2, 6, 7, 8, 9, 13        |
| 11  | Impact of external organic matter sources on nutrient, sediment and water transport from cropland into streams                               | R. Hood-Nowotny      | WEF            | Con, Met          | 12, 14, 15, 9            |
| 12  | Impacts of bioenergy crops used as riparian buffer strips on stream ecosystems' functioning                                                  | Gabriele Weigelhofer | WEF            | Met, Con, Vul     | 11, 14                   |
| 13  | Sustainable hydropower & and the water-food-energy nexus in multi-stressed IRLs: Challenges and solutions for science & management           | Stefan Schmutz       | WEF            | Vul, Con, Gov     | 14, 15, 7, 10            |
| 14  | Integrated modelling of farmers' climate change impact and adaptation appraisal to inform regional human river nexus management              | Erwin Schmid         | WEF            | Gov, Vul, Met     | 4, 11, 12, 15            |
| 15  | Strategic planning approaches to territorial governance of industrialized riverine landscapes                                                | Gernot Stöglehner    | WEF            | Gov, Vul, Met     | 11, 13, 14, 6            |

### ***Extreme events***

IRLs are complex systems shaped by the hydrological cycle and altered by human interventions. To understand the coupling of social with natural systems in IRLs, extreme events are of particular importance <sup>[3,4]</sup>. They pose manifold threats and **disturbances to ecosystems and societies**, causing inundation of settlements, infrastructure or agricultural land, but also lead to a deterioration of water quality or lack of water resources for ecosystems and society. The **magnitude and frequency** of hydrometeorological extremes is expected to increase regionally due to **global change** <sup>[49]</sup> of hydrological, biogeochemical, and biological dynamics in river systems. These processes are challenging the integrity and functioning of ecosystems, thereby restricting the provision of ecosystem services for society <sup>[50]</sup>. These restrictions can also lead to long-term natural and societal transformation processes. Therefore, determining short- and long-term changes in connectivity and water transit times is of importance, not only for assessing hydrological aspects but also the transport of pollutants affecting water quality.

Social systems have a long history of attempts to defend themselves against hydrological hazards, but adaptation **strategies** have – especially since the onset of industrialization – focused on protection, rather than on resilient system development, using technical or natural measures as the main instrument. While the costs of technical flood protection measures have been studied particularly intense, the material resources flow embedded in hydraulic structures and extreme events as drivers of construction have not yet investigated jointly. However, the **societal resource demand** is expanding and thus the **material flows** are altered. This will lead to societal interactions and feedbacks with nature and the hydrological cycle causing even more complex water management problems and raising the following research questions: What are the material flows and resource demands accumulated in hydraulic structures against floods and low flow in specific rivers? How do these technical measures impact flow and sediment transport? How can the improved morphology of a river decrease flood risks and damage? (Topics 1 and 2 addressing **metabolism and governance**). Further, we need to study the hydrological, biogeochemical, and biological changes in IRL that challenge the integrity and functioning of these ecosystems during extreme events: What are the **temporal dynamics** of drought events and how do they affect the connectivity, ecosystem functions and services of aquatic ecosystems? What are the **short-term and long-term** impacts of extreme events for IRL and which **indicators** can be used for assessing the vulnerability of SEHS (Topics 3, 4, and 5 addressing **connectivity, metabolism and vulnerability**). With these questions, HR21 will advance the understanding of **impact chains and feedback processes** of hydrological extremes on water quantity, quality, ecology, and society.

### ***Urbanization and infrastructure***

The current pronounced global trend of urbanization is one of the most important drivers of global environmental change <sup>[51]</sup>. It has significantly contributed to the development of IRL via the **built environment**. We find more and more residential, commercial, and industrial areas, roads, bridges, and other **transport** facilities in (former) floodplains. The river itself has become a waterway with ports; facilities for water supply, wastewater treatment and disposal divert and convert the flow of water and impact water quality. **Energy** is directly produced via **hydropower** or indirectly via the water-cooling of caloric nuclear power plants or other industrial facilities, and also affecting IRL via a grid of crossing power lines. Additionally, **flood protection**, settlements, **irrigation**, and **drainage** for agriculture interact with the aquatic ecosystems. In all these **infrastructures**, the historical process of **industrialization** has manifested itself.



Many of these industrial riverine **infrastructures are critical** <sup>[52]</sup> to modern human societies. Cities do not only directly affect river stretches in their own territory, they also transform distant riverine landscapes into **urban hinterlands** from which they draw **resources** and where they deposit their **wastes and emissions**. Such material and physical flows between cities and (riverine) hinterlands are investigated by **urban metabolism** research <sup>[53]</sup>. In Europe, aquatic ecosystems have been **colonized** extensively and profoundly since the 19th century and become **SEHS** via infrastructure linking rivers with the increasingly **fossil energy-based metabolism** of society <sup>[54,55]</sup>.

A successful sustainability transformation of IRL will depend on when and how **urban form and infrastructure** in floodplains will change, be further used, abandoned, or newly built <sup>[56]</sup>. Governing sustainable urbanisation and infrastructure development calls for basic integrated research to understand **conflicting goals and trade-offs**. A more **circular economy** would close at least some of the loops of the high-input-high-output metabolism of industrial societies <sup>[57]</sup> which are deeply altering river systems. The systematic integration of (urban) social with ecological metabolism studies <sup>[58]</sup> is a research frontier which will be addressed in this RF with the following research questions:

What are the consequences of infrastructure and industrial development for **instream processes**, at the level of whole river networks? How do ecosystem functioning and **biodiversity** patterns respond to specific infrastructure elements in IRL (Topic 7 addressing vulnerability and governance, Topic 8 addressing metabolism and connectivity)? To what extent has the industrialization of water courses decreased the **circularity** in **urban material flows** (Topic 9, addressing metabolism and governance) and how do these changes in urban metabolism impact **stream metabolism** (Topic 10, metabolism and connectivity)? What roles do cities play as a sink of nutrients and other materials from the floodplains (e.g., via consumption of biomass, gravel as **construction material**). How did this change during industrialization and what can we learn from this history for a more efficient and ecological organisation of such nutrient flows between cities and rivers? What is the potential of **former fluvial landscape elements** to be used as a re-activated '**green**' **infrastructure** (Topic 6 addressing metabolism and governance)?

### ***Water-energy-food-nexus (WEF)***

The water-energy-food nexus (WEF) is central to the sustainable transformation of IRL and their ecosystem perspective given the **increasing resource demand** of a growing population and further environmental and economic aspects of water, food, and energy security <sup>[59]</sup>. In the context of SEHS, the WEF provides "**cross-sectoral and dynamic perspective**" <sup>[61]</sup> and deals with the complex, often conflicting interrelations of water, energy and land resources, which exert multiple stressors on IRLs, mainly related to water quality and quantity and biodiversity, increasing agricultural and forest production, urbanization, recreational use, and energy generation both within the aquatic system (hydropower) and in the catchment (e.g., wind and solar energy, biofuel crops) <sup>[60]</sup>. Applying the concept of SEHS, our approach connects WEF nexus research <sup>[62]</sup> with research questions specific for IRL. As the WEF **integrates water and land management** as well as governance approaches across sectors and scales, it aims, inter alia, at higher resource efficiency, the reduction of trade-offs and an increase in synergies for more **policy coherence** <sup>[63]</sup>.

WEF research in IRL is based on the interactions between societal uses, demands, decisions, and adaptations, matter and energy flows linking catchments and riverine landscapes, and their ecological state. Climate change adds additional stress on riparian ecosystems and potentially increases land use demand due to losses in agricultural productivity <sup>[64]</sup>, and **climate mitigation strategies** also add additional pressures to riverine landscapes via the **energy transition**: on the one hand, **bioenergy** crops might be suitable to grow in flood prone areas, but add additional land use pressure to riverine landscapes.

These issues raise the following research questions addressing connectivity and metabolism: Which critical **flows from the terrestrial ecosystems** to the surface water and groundwater can be identified, and to what extent do they influence the capacity of riparian buffers? (Topic 11, addressing connectivity). How do different plants and crops used in riparian buffer systems affect the metabolism, organic matter decomposition, and dissolved and particulate organic matter cycling in **agricultural streams**? (Topic 12, addressing metabolism, in close cooperation with topic 11).

On the other hand, **hydropower** developments at different scales are often justified by **climate mitigation** policies, together with an increasing demand for food and fibre they pose multiple stressors to aquatic ecosystems. What are **dominant stressors** related to the energy transition impacting SEHS and what are effective mitigation, restoration, and conservation measures? (Topic 13, vulnerability and governance). In order to better understand management options and governance structures to transform IRL, it is important to understand **behavior of key stakeholders** in the WEF, especially farmers. This raises the question, dealing with vulnerability, metabolism and governance: What are farmers' behavioral intentions towards climate change adaptation and to what extent are they considered **consistent with WEF management** demands? (Topic 14, addressing governance) Finally, this RF looks at the capacity of existing **governance structures** in IRL that presumably are not yet suitable to manage these systems with regard to the WEF across sectors and scales. Which benefits and trade-offs in **strategic planning** of the WEF can be identified, and which governance structures are needed to manage IRL in a sustainable way? (Topic 15, addressing governance).

### 1.5. Gender equity

HR21 supports the **EU Gender Action Plan 2021-2025**. Gender issues permeate many of the topics of HR21 and will include gender questions and consideration of Gender Mainstreaming approaches. River engineering itself is a gendered practice, partly because of the gender bias in the engineering professions. **Participation in decision-making** processes related to IRL is also a **gendered issue** and will be recognized in all PhD topics related to stakeholder interactions, workshops, and interviews. **HR21 targets 50% female PhD students**. In addition, HR21 seeks to increase the number of female PIs by **promoting female senior scientists** (four out of six **associated faculty members**).

### 1.6. Ethical aspects

HR21 adhere to the rules of sound scientific practice as established by the Austrian Agency for Research Integrity and applies the **European Charter for Researchers**. This includes best-practice compliance to the General Data Protection Regulation and best ethical practice when handling sensitive data. Preservation of personal sphere, professional role and anonymization of outputs will also be considered, as well as any **relevant regulations** concerning ethical issues.

## Annex 1: List of references

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